

Software Requirement Specification (SRS)

1. Introduction

1.1 Project Title: NextGen Campus AI

1.2 Project Overview:

NextGen Campus AI is an AI-powered web-based platform designed to assist students, teachers, and administrators in managing academic activities efficiently. The system combines traditional campus management functionalities with artificial intelligence to provide academic assistance, routine management, assignment tracking, quiz generation, study planning, note summarization, and intelligent question answering.

The platform aims to serve as a centralized digital assistant that improves communication, organization, and learning experiences within a university environment.

1.3 Problem Statement:

University students often face difficulties in managing academic information and maintaining productivity due to the scattered nature of campus resources. Students may struggle to track class schedules, examination routines, assignments, class tests etc. Additionally, obtaining quick academic support and personalized study guidance can be challenging.

Existing academic management systems primarily provide static information and lack intelligent assistance. As a result, students spend considerable time searching for information and organizing their academic activities manually.

To address these challenges, the NextGen Campus AI proposes an integrated platform that combines campus management services with artificial intelligence to provide instant academic support and personalized learning assistance.

1.4 Objectives:

The main objectives of the NextGen Campus AI are:

- To provide a centralized platform for academic information management.
- To help students' access class routines, exam schedules, assignments, and notices efficiently.
- To enable teachers to manage assignments, class tests, and academic announcements.
- To assist administrators in managing users and academic information.
- To provide AI-powered academic assistance through intelligent question answering.
- To help students understand topics, summarize study materials for faster learning, and automatically generate quizzes on selected topics.
- To improve overall student productivity and academic performance.

1.5 Scope of the Project:

The NextGen Campus AI will provide academic management and AI-assisted learning support through a web-based platform.

The system will support three primary user roles: Administrator, Teacher, and Student.

- Administrator Scope:
 - ✓ Manages user accounts (student, teacher).
 - ✓ Manages courses academic information.
 - ✓ Manages class schedules and examination schedules.
 - ✓ Monitor overall system activities.
- Teacher Scope:
 - ✓ Create and manage assignments.
 - ✓ Publish class test notice.
 - ✓ Upload course materials and resources.
 - ✓ Generate quizzes using AI assistance.
- Student Scope:
 - ✓ View class routines and examination schedules.
 - ✓ Access assignments and notices.
 - ✓ Interact with AI-powered academic assistant.
 - ✓ Generate quizzes for practice.
 - ✓ Summarize study notes and lecture materials.
- AI Scope:
 - ✓ Academic question answering.
 - ✓ Study plan generation.
 - ✓ Assignment assistance.
 - ✓ Quiz generation.
 - ✓ Note summarization.

The system will be accessible through modern web browsers and will integrate with AI services such as the Gemini API or other suitable APIs to provide intelligent learning support.

2. Functional Requirements

Functional requirements describe the specific functions and services that the NextGen Campus AI system must provide.

2.1 Authentication Module:

FR-01: Account Creation

The system shall allow administrators to create and manage student and teacher accounts.

FR-02: User Login

The system shall allow registered users to log in using their credentials.

FR-03: User Logout

The system shall allow users to securely log out from the system.

FR-04: Role-Based Access Control

The system shall provide access permissions based on user roles (Administrator, Teacher, and Student).

FR-05: Password Management

The system shall allow users to change their passwords securely.

2.2 Administrator Module:

FR-06: Manage Student Accounts

The system shall allow administrators to create, update, view, and delete student accounts.

FR-07: Manage Teacher Accounts

The system shall allow administrators to create, update, view, and delete teacher accounts.

FR-08: Manage Courses

The system shall allow administrators to create, update, and manage course information.

FR-09: Manage Class Routines

The system shall allow administrators to create, update, and publish class schedules.

FR-10: Manage Examination Routines

The system shall allow administrators to create, update, and publish examination schedules.

FR-11: Manage Academic Notices

The system shall allow administrators to create and publish important academic notices.

FR-12: Monitor System Information

The system shall allow administrators to view system statistics and academic information.

2.3 Teacher Module:**FR-13: Teacher Login Access**

The system shall allow teachers to access teacher-specific functionalities after authentication.

FR-14: Create Assignments

The system shall allow teachers to create and publish assignments.

FR-15: Manage Assignments

The system shall allow teachers to edit, update, and delete assignments.

FR-16: Create CT Notices

The system shall allow teachers to create and publish class test notices.

FR-17: Upload Course Materials

The system shall allow teachers to upload lecture notes, documents, and academic resources.

FR-18: View Student Information

The system shall allow teachers to view relevant student academic information.

FR-19: Generate AI-Based Quizzes

The system shall allow teachers to generate quizzes automatically using AI services.

2.4 Student Module:

FR-20: Student Login Access

The system shall allow students to access student-specific functionalities after authentication.

FR-21: View Dashboard

The system shall provide students with a dashboard displaying academic information and updates.

FR-22: View Class Routines

The system shall allow students to view class schedules.

FR-23: View Examination Routines

The system shall allow students to view examination schedules.

FR-24: View Assignments

The system shall allow students to access assignment details and deadlines.

FR-25: View Academic Notices

The system shall allow students to view academic announcements and notices.

FR-26: Access Course Materials

The system shall allow students to view and download uploaded academic resources.

2.5 AI Features Module:

FR-27: AI Chat Assistant

The system shall provide an AI chatbot capable of answering academic questions submitted by users.

FR-28: AI Study Planner

The system shall generate personalized study plans based on user-provided subjects and study duration.

FR-29: AI Assignment Helper

The system shall assist students in understanding assignment requirements and academic topics.

FR-30: AI Quiz Generator

The system shall generate quizzes from selected topics using AI services.

FR-31: AI Note Summarizer

The system shall summarize lecture notes and study materials provided by users.

FR-32: AI Response Generation

The system shall communicate with external AI services to generate intelligent responses.

2.6 Functional Requirements Summary:

Module	Requirement Count
Authentication	5
Administrator	7
Teacher	7
Student	7
AI Features	6
Total	32 Requirements

3. Non-Functional Requirements

Non-functional requirements specify the quality attributes, performance expectations, security measures, and operational constraints of the NextGen Campus AI.

3.1 Performance Requirements:**NFR-01: Page Load Time**

The system should load normal web pages within **3 seconds** under standard network conditions.

NFR-02: API Response Time

The system should provide responses for standard requests within **2 seconds**, excluding AI-generated responses.

NFR-03: AI Response Time

The system should generate AI responses within a reasonable time, typically **5–10 seconds**, depending on the complexity of the request and external AI service availability.

NFR-04: Concurrent Users

The system should support multiple users accessing the platform simultaneously without significant performance degradation.

3.2 Security Requirements:**NFR-05: Secure Authentication**

The system shall require authentication before granting access to protected resources.

NFR-06: Password Encryption

User passwords shall be stored in encrypted or hashed form within the database.

NFR-07: Role-Based Authorization

Users shall only be able to access features permitted by their assigned role.

NFR-08: Data Protection

The system shall protect user data from unauthorized access and modification.

NFR-09: Secure API Communication

Communication between the application and AI services should be conducted using secure protocols such as HTTPS.

3.3 Usability Requirements:

NFR-10: User-Friendly Interface

The system shall provide a simple and intuitive user interface for all users.

NFR-11: Responsive Design

The system shall support desktop, tablet, and mobile devices through responsive web design.

NFR-12: Easy Navigation

Users should be able to access major functionalities with minimal navigation effort.

NFR-13: Consistent User Experience

The system shall maintain a consistent layout and design across all modules.

3.4 Reliability Requirements:

NFR-14: System Availability

The system should be available whenever users need access, except during scheduled maintenance periods.

NFR-15: Error Handling

The system shall provide meaningful error messages when failures occur.

NFR-16: Data Integrity

The system shall maintain accurate and consistent data during all operations.

NFR-17: Backup and Recovery

The system should support database backup and recovery procedures to minimize data loss.

3.5 Scalability Requirements:

NFR-18: User Scalability

The system should be designed to accommodate future growth in the number of users.

NFR-19: Feature Scalability

The system architecture should allow new modules and functionalities to be added with minimal modification.

NFR-20: AI Service Integration Scalability

The system should support replacing or upgrading AI service providers when required.

3.6 Compatibility Requirements:

NFR-21: Browser Compatibility

The system should function correctly on modern web browsers such as:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

NFR-22: Cross-Platform Accessibility

The system should be accessible from Windows, Android, and other commonly used operating systems through a web browser.

3.7 Maintainability Requirements:

NFR-23: Modular Design

The system should follow a modular architecture to simplify maintenance and future development.

NFR-24: Code Maintainability

The source code should be organized and documented to support future modifications and debugging.

NFR-25: Version Control Support

The project source code shall be maintained using a version control system such as Git.

3.8 Non-Functional Requirements Summary:

Category	Requirements
Performance	4
Security	5
Usability	4
Reliability	4
Scalability	3
Compatibility	2
Maintainability	3
Total	25 Requirements

4. User Roles

The NextGen Campus AI stant system consists of three primary user roles: **Administrator**, **Teacher**, and **Student**. Each role has different responsibilities and access permissions within the system.

4.1 Administrator:

The Administrator is responsible for managing the overall system and academic information.

Responsibilities:

- Manage student and teacher accounts.
- Manage course information.
- Manage class routines and examination routines.
- Publish academic notices.
- Monitor system activities.
- Maintain data accuracy and consistency.

Permissions:

- Full access to all administrative features.
- Create, update, view, and delete system data.
- Manage users and academic resources.

4.2 Teacher:

The Teacher is responsible for managing course-related activities and supporting student learning.

Responsibilities:

- Create and manage assignments.
- Publish class test (CT) notices.
- Upload course materials and academic resources.
- Generate quizzes using AI features.
- View relevant student academic information.

Permissions:

- Access teacher dashboard.
- Manage assignments and notices.
- Upload and manage learning materials.
- Use AI quiz generation features.

4.3 Student:

The Student is the primary beneficiary of the NextGen Campus AI system and uses the platform for academic support and information access.

Responsibilities:

- View academic information regularly.
- Track assignments and deadlines.
- Use AI learning assistance responsibly.
- Maintain account information.

Permissions:

- Access student dashboard.
- View class routines.
- View examination schedules.
- Access assignments and notices.
- View course materials.
- Interact with the AI Chat Assistant.
- Generate study plans.
- Use AI Assignment Helper.
- Generate quizzes.
- Summarize lecture notes and study materials.

5. Use Cases

5.1 User Login

- **Actor:** Administrator / teacher / student
- **Action:** The user enters their username/email and password and submits the login form
- **System Response:** The system validates the credentials and grants access to the appropriate dashboard based on the user's role.

5.2 Manage Users

- **Actor:** Administrator
- **Action:** The administrator creates, updates, views, or deletes student and teacher accounts.
- **System Response:** The system processes the requested operation and updates the user database accordingly.

5.3 Manage Assignments

- **Actor:** Teacher
- **Action:** The teacher creates, updates or removes assignments and publishes them for students.
- **System Response:** The system stores the assignments information and makes it available to students.

5.4 View Academic Information

- **Actor:** Student
- **Action:** The student accesses the dashboard to view class routines, examination schedules, assignments, notices and course materials.
- **System Response:** The system retrieves and displays the requested academic information.

5.5 AI Chat Assistant

- **Actor:** Student
- **Action:** The student submits an academic question through AI chat interface.
- **System Response:** The system sends the query to the AI service and returns an intelligent response to the student.

5.6 AI Quiz Generator

- **Actor:** Teacher / student
- **Action:** The user enters a topic and requests quiz generation.
- **System Response:** The system uses AI services to generate quiz questions and displays them to the user.

5.7 AI Study Planner

- **Actor:** Student
- **Action:** The student provides subjects, study goals or study duration and requests a study plan.
- **System Response:** The system generates a personalized study schedule using AI and displays it to the student.

5.8 AI Note Summarizer

- **Actor:** Student
- **Action:** The student uploads or provides lecture notes and requests a summary.
- **System Response:** The system analyzes the content and generates a concise summary using AI.